PLACEMENT FOCUSED

DSA ROADMAP

FOR PLACEMENTS - JAVA

A practical learning path for 1st-round coding tests and 2nd-round coding / technical interviews.

THE LEARNING LOOP

Learn concept → Solve easy → Solve medium → See pattern → Timed practice → Explain

STUDENT EDITION

Start with the highest-value interview topics.

Build confidence before moving to Graphs and Dynamic Programming.

Your Complete DSA Journey

Follow the sequence. Do not rush into advanced topics before the foundations are comfortable.

1. Java Problem Solving
2. Arrays
3. Strings
4. Time & Space Complexity
5. Searching
6. Sorting
7. Hashing
8. Two Pointers
9. Sliding Window
10. Prefix Sum
11. Linked List
12. Stack
13. Queue / Deque
14. Recursion
15. Binary Search
16. Trees
17. BST
18. Heap / Priority Queue
19. Greedy
20. Graphs
21. Dynamic Programming
22. Advanced Topics

RULE OF PROGRESSION

Foundation -> Patterns -> Core DSA -> Advanced Placement DSA -> Interview Practice

Stage 1 - Build the Foundation

Java basics, complexity thinking, and the most important first DSA topic.

LEVEL 0 Problem-Solving Foundation

START HERE

LEARN

- Be comfortable with variables, if/else, loops, methods, arrays, strings, and basic Java Collections.
- Collections needed: ArrayList, HashMap, HashSet, Queue, Deque, PriorityQueue.
- Understand O(1), O(log n), O(n), O(n log n), O(n^2), and O(2^n).

PATTERNS / PROBLEMS

- Do not just memorize complexity. Understand why an algorithm has that complexity.

TARGET **Strong basics before DSA**

LEVEL 1 Arrays

5 / 5 PRIORITY

This should be your first major DSA topic.

LEARN

- Traversal
- Insertion
- Deletion
- Searching
- Maximum / minimum
- Sum / average
- Frequency
- Reverse
- Second largest
- Duplicates
- Missing number
- Sorting
- Rotation
- Merge arrays

PATTERNS / PROBLEMS

- Two pointers
- Sliding window
- Prefix sum
- Kadane's algorithm

TARGET **30-40 problems**

Stage 2 - Strings & Search

Master common string operations, then make binary search a reflex.

LEVEL 2 Strings

5 / 5 PRIORITY

LEARN

- Character traversal
- charAt()
- StringBuilder
- Reverse
- Palindrome
- Frequency
- Anagram
- Duplicate characters
- Substrings
- String comparison

PATTERNS / PROBLEMS

- Two pointers
- Hashing
- Sliding window

TARGET 20-25 problems

LEVEL 3 Searching

5 / 5 PRIORITY

LEARN

- Start with Linear Search.
- Master Binary Search and understand low, mid, high.

PATTERNS / PROBLEMS

- Search element
- First occurrence
- Last occurrence
- Lower bound
- Upper bound
- Search insert position
- Square root
- Search rotated array
- Find minimum in rotated array

TARGET 15-20 problems

Stage 3 - Sorting & Hashing

Learn how data is ordered, then use HashMap / HashSet to cut unnecessary work.

LEVEL 4 Sorting

4 / 5 PRIORITY

LEARN

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort

PATTERNS / PROBLEMS

- For interviews know: how it works, time complexity, space complexity, and when to use it.

TARGET 10-15 problems + implementation practice

LEVEL 5 Hashing

5 / 5 PRIORITY

Extremely important for placement coding.

LEARN

- Use HashMap and HashSet.

PATTERNS / PROBLEMS

- Frequency counting
- Duplicate detection
- Two Sum
- Common elements
- First non-repeating character
- Count occurrences
- Grouping
- Pair problems

TARGET 15-20 problems

Stage 4 - High-Value Problem-Solving Patterns

These are patterns, not separate data structures. Learn when to recognize them.

LEVEL 6 Two Pointers

5 / 5 PRIORITY

LEARN

- Use left and right pointers, often moving inward or forward.

PATTERNS / PROBLEMS

- Sorted arrays
- Pair sum
- Remove duplicates
- Palindrome
- Reverse
- Container problems

TARGET 10-15 problems

LEVEL 7 Sliding Window

5 / 5 PRIORITY

LEARN

- Fixed window
- Variable window using left and right

PATTERNS / PROBLEMS

- Maximum sum subarray
- Longest substring
- Minimum window
- Longest subarray satisfying condition

TARGET 10-15 problems

LEVEL 8 Prefix Sum

4 / 5 PRIORITY

LEARN

- prefix[i] = prefix[i-1] + arr[i]

PATTERNS / PROBLEMS

- Range sum
- Subarray sum
- Frequency / range problems
- Subarray-related optimization

TARGET 8-10 problems

Stage 5 - Core Linear Data Structures

Now move from array-based thinking to nodes, LIFO, FIFO, and deque patterns.

LEVEL 9 Linked List

5 / 5 PRIORITY

LEARN

- Create list
- Traverse
- Insert at beginning
- Insert at end
- Delete
- Search
- Reverse

PATTERNS / PROBLEMS

- Find middle
- Reverse linked list
- Detect cycle
- Remove cycle
- Merge two sorted lists
- Find intersection
- Remove nth node

TARGET 15-20 problems

LEVEL 10 Stack

5 / 5 PRIORITY

LEARN

- Understand LIFO
- Push
- Pop
- Peek
- Empty check
- Use Deque<Integer> stack = new ArrayDeque<>();

PATTERNS / PROBLEMS

- Balanced parentheses
- Reverse string
- Next greater element
- Previous greater element
- Min stack
- Remove adjacent duplicates

TARGET 10-15 problems

LEVEL 11 Queue & Deque

4 / 5 PRIORITY

LEARN

- Understand FIFO
- Queue
- Circular queue
- Deque
- Priority queue basics

PATTERNS / PROBLEMS

- Queue implementation
- Generate binary numbers
- Sliding-window maximum
- First non-repeating character

TARGET 8-12 problems

Stage 6 - Recursion & Trees

Recursion becomes practical when tree traversal and backtracking start to click.

LEVEL 12 Recursion

4 / 5 PRIORITY

LEARN

- Base Case -> Recursive Call -> Return
- Factorial
- Fibonacci
- Sum of numbers
- Reverse string
- Array traversal
- Power
- Digit problems

PATTERNS / PROBLEMS

- Backtracking: subsets
- Permutations
- Combination problems
- Basic maze problems

TARGET 15-20 problems

LEVEL 13 Trees

5 / 5 PRIORITY

LEARN

- Binary Tree
- Node
- Height
- Number of nodes
- Leaf nodes
- Search

PATTERNS / PROBLEMS

- Inorder
- Preorder
- Postorder
- Level Order
- Preorder = Root Left Right
- Inorder = Left Root Right
- Postorder = Left Right Root

TARGET 15-20 problems

Stage 7 - Ordered Trees, Heaps & Greedy

Add priority-based structures and learn when a locally best decision works.

LEVEL 14 Binary Search Tree

4 / 5 PRIORITY

LEARN

- Insert
- Search
- Delete
- Minimum
- Maximum
- Inorder traversal
- Validate BST
- Lowest Common Ancestor

TARGET 10-15 problems

LEVEL 15 Heap / Priority Queue

4 / 5 PRIORITY

LEARN

- Java: PriorityQueue<Integer>
- Min Heap: smallest element first
- Max Heap: largest element first

PATTERNS / PROBLEMS

- Kth largest
- Kth smallest
- Top K elements
- Merge sorted arrays
- Priority scheduling

TARGET 10-15 problems

LEVEL 16 Greedy

3 / 5 PRIORITY

You do not need dozens initially.

LEARN

- Make the best choice available at the current step.

PATTERNS / PROBLEMS

- Activity selection
- Fractional knapsack
- Minimum platforms
- Jump Game

TARGET 8-12 problems

Stage 8 - Advanced Placement DSA

Start these only after your core topics are comfortable.

LEVEL 17 Graphs

4 / 5 PRIORITY

LEARN

- Start after trees.
- Understand vertices and edges.
- Representations: Adjacency Matrix and Adjacency List.

PATTERNS / PROBLEMS

- BFS
- DFS
- Connected components
- Cycle detection
- Shortest path basics
- Topological sort
- Dijkstra basics

TARGET **15-20 problems**

LEVEL 18 Dynamic Programming

5 / 5 PRIORITY

Advanced DP mastery is not required before applying to most entry-level roles.

LEARN

- Do not start DP early. Become strong in Arrays + Recursion + Trees + Hashing first.

PATTERNS / PROBLEMS

- Level 1: Fibonacci, Climbing stairs, House robber
- Level 2: 0/1 Knapsack, Subset sum, Coin change
- Level 3: Longest Common Subsequence, Longest Increasing Subsequence, Grid DP

TARGET **15-20 carefully selected problems**

Placement Priority Map

Spend your time where interview return-on-effort is highest.

Topic	Priority
Arrays	5/5
Strings	5/5
Hashing	5/5
Searching	5/5
Sorting	4/5
Two Pointers	5/5
Sliding Window	5/5
Linked List	4/5
Stack	4/5
Queue	3/5
Recursion	4/5
Trees	4/5
BST	3/5
Heap	3/5
Graph	3/5
Greedy	3/5
DP	4/5
Advanced DSA	1/5

MOST IMPORTANT FOCUS

Arrays -> Strings -> Hashing -> Two Pointers -> Sliding Window -> Searching -> Sorting

Then: Linked List -> Stack -> Trees

How Many Problems Should You Solve?

Quality beats blindly chasing 500-1000 problems.

PHASE 1 — 50 Java basic problems

↓

PHASE 2 — 50 Arrays + Strings

↓

PHASE 3 — 50 core DSA

↓

PHASE 4 — 50 mixed Easy / Medium

↓

FINAL — 150-200 quality DSA problems

USEFUL TOPIC DISTRIBUTION

Arrays	35	Strings	20	Hashing	20
Searching	15	Sorting	10	Two Pointers	10
Sliding Window	10	Linked List	15	Stack/Queue	15
Recursion	10	Trees	15	Heap	8
Graph	10	Greedy	8	DP	10

These numbers are targets, not strict quotas. The roadmap totals about 211 problems.

The Most Important Rule

For every problem, follow the same deliberate practice process.

1. **Read problem**
2. **Understand input / output**
3. **Think of brute force**
4. **Write brute-force solution**
5. **Analyze complexity**
6. **Find better approach**
7. **Code optimized solution**
8. **Test edge cases**
9. **Explain solution without looking**

WHY THIS WORKS

Consistent problem analysis builds DSA skill faster than only watching solution videos.

The goal is to recognize patterns and explain your reasoning under interview pressure.

Final Placement Roadmap

A simple view of where you are now and what comes next.

BEGINNER

- Java
- Complexity
- Arrays
- Strings

INTERMEDIATE

- Searching
- Sorting
- Hashing
- Two Pointers
- Sliding Window
- Prefix Sum

CORE DSA

- Linked List
- Stack
- Queue
- Recursion
- Trees
- BST

ADVANCED PLACEMENT DSA

- Heap
- Greedy
- Graph
- Dynamic Programming

FINAL STAGE

- Topic-wise Problems
- Mixed Problems
- Company Problems
- Timed Coding Tests
- Mock Interviews
- REAL INTERVIEW

YOUR HIGH-PRIORITY COMFORT ZONE

Arrays -> Strings -> Hashing -> Two Pointers -> Sliding Window -> Searching -> Sorting -> Linked List -> Stack -> Trees